What Does Our Anomaly Detector Detect? Plain Summary of the Corrected Evidence

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Abstract

The label problem is fixed for the main CERT data and score evaluations. The main language-model adapters remain usable. Some internal experiments and comparison models still need repair because labels influenced their fitting or selection. The clearest complete finding is that scoring behavior targets improves r4.2 ranking while the model still sees profile information. That improvement does not yet tell us which internal mechanism caused it.

1 The core question

We turn one employee's day into text containing profile fields and session activity. A language model adapted only on non-insider users assigns an anomaly score: the average surprise of the next tokens. We want to know whether its score reflects suspicious behavior, unusual profiles, or exposure during training.

These are three different experiments:

1. Change which token losses enter the score, leaving the input and model fixed.
2. Change profile input or adaptation, and measure the resulting predictions.
3. Edit an internal feature and measure the resulting change in output.

Success in the first does not establish the mechanism in the third.

2 What the label correction changed

The old calendar join assigned some incident windows to the wrong observed days. The corrected recorded-date target has these populations:

Dataset	Observed days	Positive days	Insiders	Changed labels
r4.2	330,295	1,362	70	407
r6.2	1,393,297	73	5	23

An incident-window day is the target; an insider's other days are *window-negative*, not verified benign. The corrected clock and every label were independently checked. Texts and user splits are unchanged. The main adapters trained on no official insiders, so this bug does not require retraining them. It does affect experiments that selected normal users or features from the faulty day labels.

3 What we can currently conclude

Evaluation population matters. The main LM's fold day ROC is 0.960 on r4.2 and 0.956 on r6.2 when many comparison users were seen in training. Restricting comparison users to those absent from adaptation

gives 0.538 and 0.583. This identifies a large evaluation gap; it does not isolate its sole cause. Nine newly included r4.2 insiders have especially poorly ranked incident-window days.

Changing the score helps on the complete r4.2 population. The separate score-decomposition pass covers all 70 insiders:

r4.2 score	Day ROC	Day AP	User ROC
Full score	0.540	0.004859	0.589
Behavior targets	0.820	0.025238	0.825
Profile targets	0.482	0.001353	0.476

Behavior-only scoring means averaging losses on behavior targets, while the model still reads the full profile. Its day-ROC gain is 0.280, with a 95% descriptive user-bootstrap interval [0.204, 0.357]. This is useful score mitigation; internal profile independence is unproven. Average precision is not the fraction of alerts that would be correct at a selected threshold. The tables report equal-insider fold means, not pooled metrics.

Other comparisons are narrower. In separate r6.2 factorial adapters, removing profiles from both training and scoring improves ranking in the audited 3B and 8B runs. There is one seed per condition and only five insiders, so this is not a general model-size law. Removing profile training targets while retaining the input does not establish an additional improvement over behavior scoring. Cached numeric temporal models also have corrected results, but history-copying and causal explanations remain open. The full reading edition gives the separate populations and numerical caveats.

4 How interpretability fits in

We compare the internal states of the frozen model with adapters on and off: $\delta = h_{\text{adapted}} - h_{\text{base}}$. A sparse autoencoder approximates these changes using a few learned directions. We select candidate directions, edit them, and compare the output change with active and random controls. We also measure where features activate in the input.

This is a way to investigate the score, not an automatic explanation. A feature at a session token can use earlier profile context. The size of an intended edit can differ from the edit actually represented in low precision. We therefore check the realized perturbation and compute unedited and edited losses in the same numerical configuration. Sparse coding and optimization ideas remain useful methodological ingredients, but we are not asserting a new mathematical theorem.

5 What is still unresolved

The corrected internal pilot completed 4,740 scored conditions on 316 receivers, but one control token missed its fixed norm-matching tolerance. Its original global gate remains failed. Separate diagnostics show sensitivity to selected edits but do not establish attack specificity. A small control-policy probe is not a replacement for a fully validated experiment.

Historical profile-swap and H6 responses can be retained for the fixed inputs. Their corrected class summaries and partner checks do not restore the original matching. Old attribution, transfer, robustness, and mechanistic replication claims remain pending wherever fitting or selection used the wrong labels.

The comparison models need separate repair: saved local-autoencoder training membership is contaminated in 94/120 r4.2 and 8/8 r6.2 bundles, and the old sequence baselines share the faulty fold construction. Their old scores do not establish that QLoRA is better. TWOS and LANL have separate unresolved issues and are not cleared replications. Small personality-novelty associations remain exploratory.

6 Where this leaves the paper

The defensible current story is that population choice and score construction matter, and that score mitigation does not settle internal information use. The interpretability method stays in the paper. The next experiments should repair the specific dependencies needed to answer the internal question, with frozen endpoints and controls. All three editions now use the same corrected result registry; the complete reading edition carries the technical details. This is a recovery working edition, not a claim that every causal result is repaired.